① $f(x) = x^2 - 3x + 1$

a)

~~[illegible]~~

$\rightarrow$ kořen v $\langle a_0, b_0 \rangle = \langle 1, 5 \rangle \rightarrow \left[\begin{array}{l} f(1) = 1^2 - 3 + 1 = (-1) \ \ominus \\ f(5) = 5^2 - 3 \cdot 5 + 1 = 11 \ \oplus \end{array} \right]$

$\rightarrow m_1 \rightarrow \frac{1+5}{2} = 3 \rightarrow f(3) = 1 \ \oplus \rightarrow$ opačné vůči $f(1) \Rightarrow$ kořen v $\langle 1; 3 \rangle$

$\rightarrow m_2 \rightarrow \frac{1+3}{2} = 2 \rightarrow f(2) = (-1) \ \ominus \rightarrow$ opačné vůči $f(3) \Rightarrow$ kořen v $\langle 2; 3 \rangle$

$\rightarrow$ odhad kořene po dvou iteracích $\hat{r} = 2{,}5$

b)

$f'(x) = 2x - 3$

$$x_{k+1} = x_k - \frac{x_k^2 - 3x_k + 1}{2x_k - 3} = \frac{x_k(2x_k - 3)}{2x_k - 3} + \frac{-x_k^2 + 3x_k - 1}{2x_k - 3} = \frac{2x_k^2 - 3x_k - x_k^2 + 3x_k - 1}{2x_k - 3} =$$

$$= \frac{x_k^2 - 1}{2x_k - 3}$$

$x_0 = 1$

$x_1 = \frac{1^2 - 1}{2 \cdot 1 - 3} = \frac{0}{-1} = 0$

$x_2 = \frac{0 - 1}{0 - 3} = 3$

c)

$\hat{r} = 3 \quad \varepsilon = \frac{1}{4}$

$f(3) = 1 \ \oplus$

$f(\frac{13}{4}) = (\frac{13}{4})^2 - 3 \cdot \frac{13}{4} + 1 = \frac{29}{16} \ \oplus$

$f(\frac{11}{4}) = (\frac{11}{4})^2 - 3 \frac{11}{4} + 1 = \frac{5}{16} \ \oplus$

$\Big\}$ $\hat{r} = 3$ není dostatečně přesný

(2)

$f(x) = x^2 - x + 1$

$f'(x) = 2x - 1$

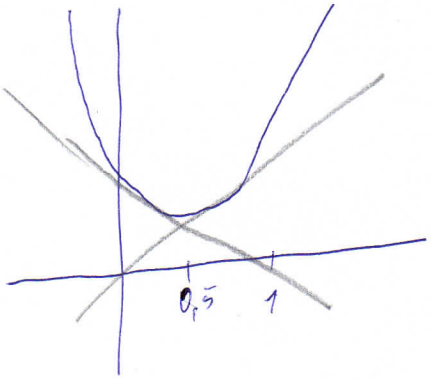
$$x_{k+1} = x_k - \frac{x_k^2 - x_k + 1}{2x_k - 1} = \frac{2x_k^2 - x_k - x_k^2 + x_k - 1}{2x_k - 1} = \frac{x_k^2 - 1}{2x_k - 1}$$

$x_0 = 2$

$$x_1 = \frac{2^2 - 1}{2 \cdot 2 - 1} = 1$$

$$x_2 = \frac{1^2 - 1}{2 \cdot 1 - 1} = \frac{0}{1} = 0$$

$$x_3 = \frac{0^2 - 1}{2 \cdot 0 - 1} = \frac{-1}{-1} = 1$$

$$x_4 = \frac{1^2 - 1}{2 \cdot 1 - 1} = \frac{0}{1} = 0$$

Kořen leží na intervalu $\langle 0; 1 \rangle$ a vzhledem ke tvaru grafu fce, je metoda tečen zacyklena na těchto bodech...